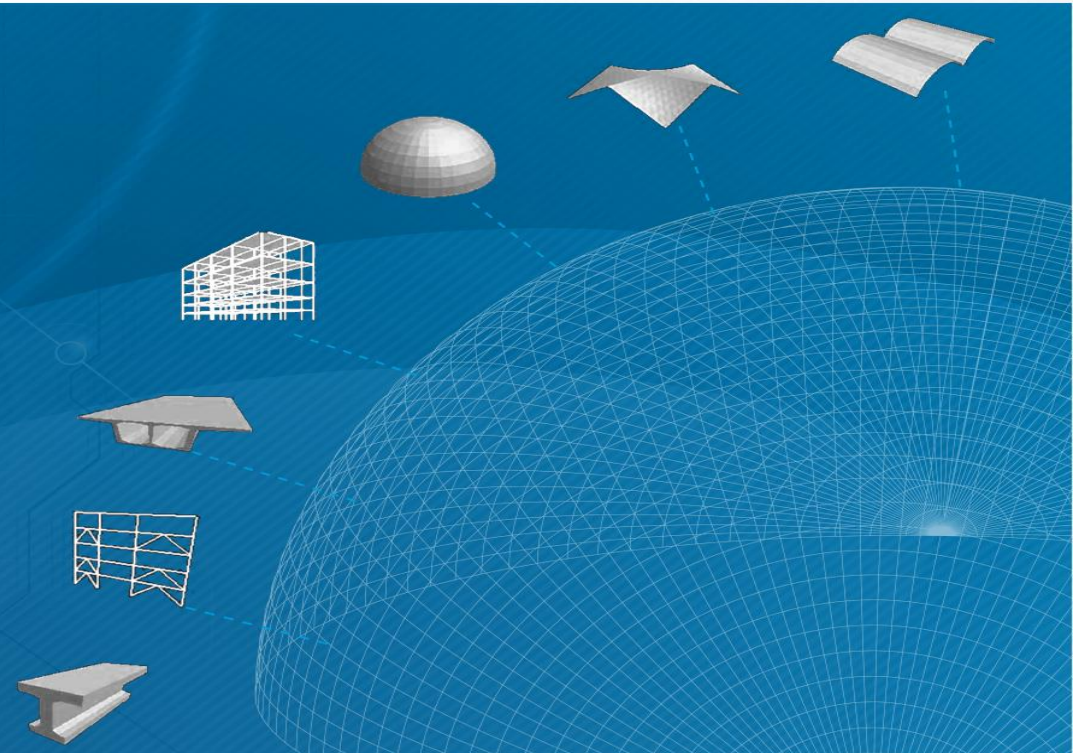
CE – 412

Design of Concrete Structures



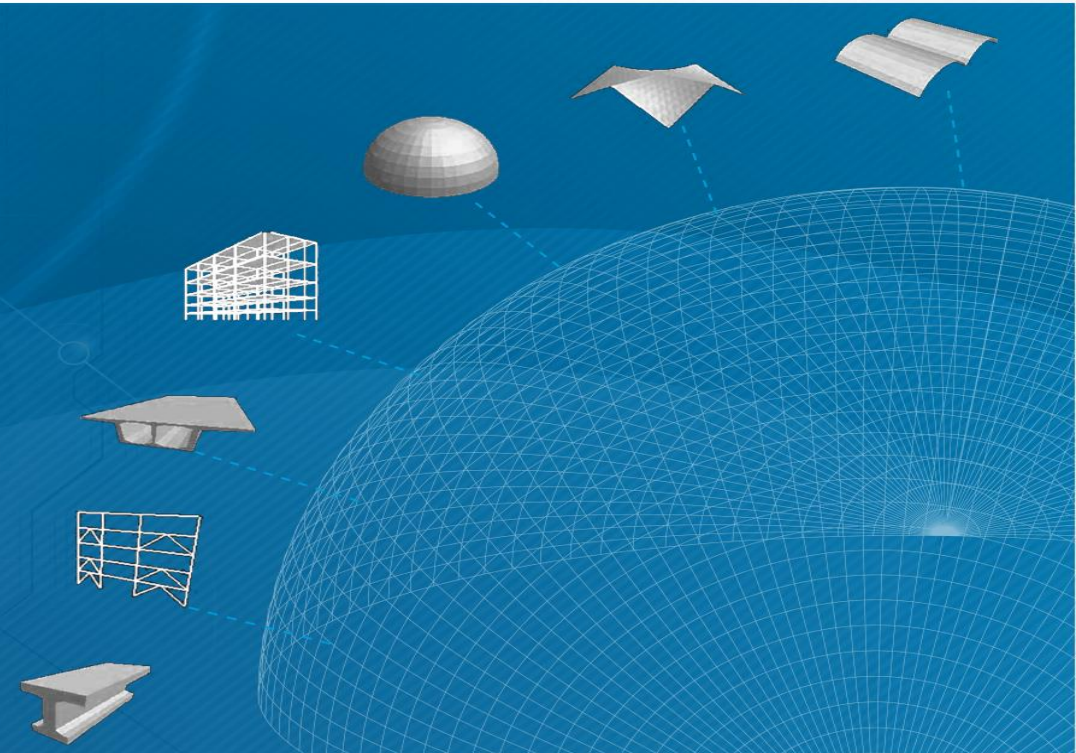
September – 2021

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NUST Institute of Civil Engineering (NICE)
National University of Sciences and Technology (NUST)
H-12 Islamabad, Pakistan
Cell: 92-334-5192533, Email: fawad@nice.nust.edu.pk

CE – 412 Design of Concrete Structures

Lecture 0:

Course Introduction



September – 2021

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General information

- Course Code — CE 412
- Course Title — Design of Concrete Structures
- Credit Hours — 3+0
- Websites — www.structurespro.info, www.fawadnajam.com
- Lecturer — Fawad Ahmed Najam
- Email — fawad@nice.nust.edu.pk
- Office — Room # 118, PG Wing (NICE)

Connected CLO & PLO's

1	Engineering Knowledge:	☐	7	Environment and Sustainability:	☐
2	Problem Analysis:	☐	8	Ethics:	☐
3	**Design/Development of Solutions:**	☑	9	Individual and Team Work:	☐
4	Investigation:	☐	10	**Communication:**	☑
5	Modern Tool Usage:	☐	11	Project Management:	☐
6	The Engineer and Society:	☐	12	Lifelong Learning:	☐

Connected CLO & PLO's

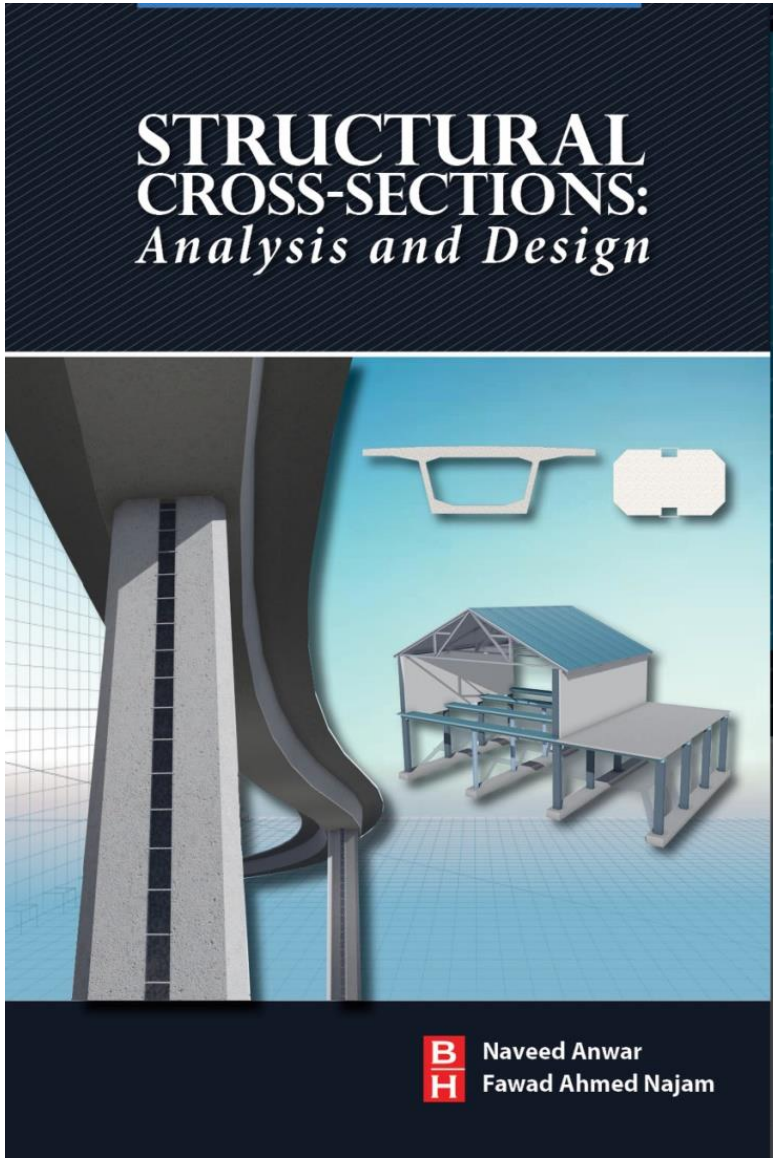
No.	CLO	Domain	Taxonomy Level	PLO
1	Design of Reinforced Concrete Structures	Cognitive	5	3
2	Development of Structural Drawings for the Designed Structures	Cognitive	5	10
3	Complex Engineering Problem – Structural Design of a Building	Cognitive	5	3

Course Outlines

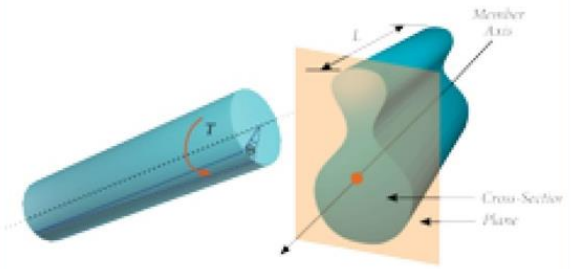
- Lecture 0: Course Introduction
- Lecture 1: Introduction to Structural Design Process
- Lecture 2: Material Behavior
- Lecture 3: Cross-sectional Behavior (Flexure) – Analysis & Design of Beams [Summary]
- Lecture 4: Cross-sectional Behavior (Shear and Torsion) – Analysis & Design of Beams
- Lecture 5: Cross-sectional Behavior (Axial-Flexural) – Analysis & Design of Short Columns [Summary]
- Lecture 6: Member Behavior – Design of Slender Columns
- Lecture 7: Design of Two-way Slabs
- Lecture 8: Design of Isolated and Combined Footings
- Lecture 9: Design of RC Shear Walls
- Lecture 10: Design of Retaining Walls

Textbook

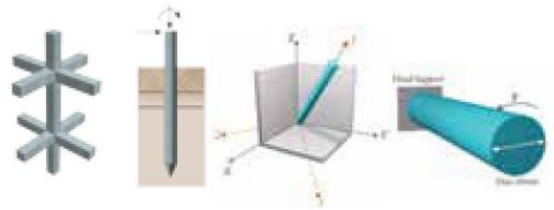
- Lecture Notes Provided by the instructor
- Anwar N., and **Najam F. A.** “Structural Cross-sections: Analysis and Design”, 1st Edition. ISBN: 9780128044438, Publisher: Elsevier, Butterworth-Heinemann, 2016 (Details: www.amazon.com/Structural-Cross-Sections-Analysis-Design/dp/0128044438)
- https://www.sciencedirect.com/book/9780128044438/structural-cross-sections



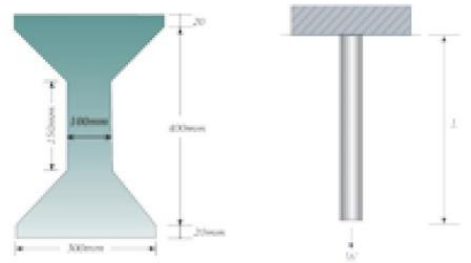
STRUCTURES AND STRUCTURAL DESIGN



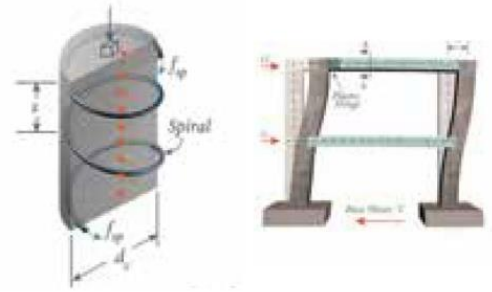
RESPONSE AND DESIGN OF COLUMN CROSS-SECTION



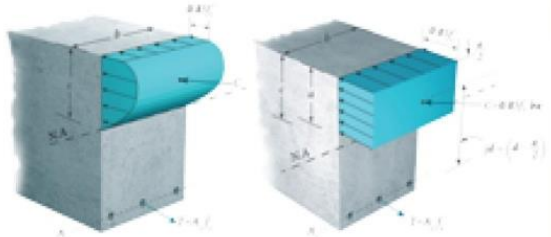
UNDERSTANDING CROSS-SECTIONS



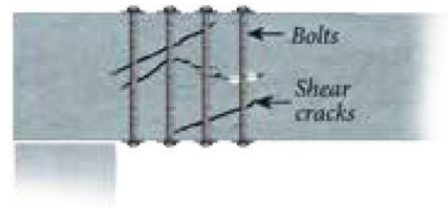
DUCTILITY OF CROSS-SECTIONS



AXIAL-FLEXUAL RESPONSE OF CROSS-SECTIONS



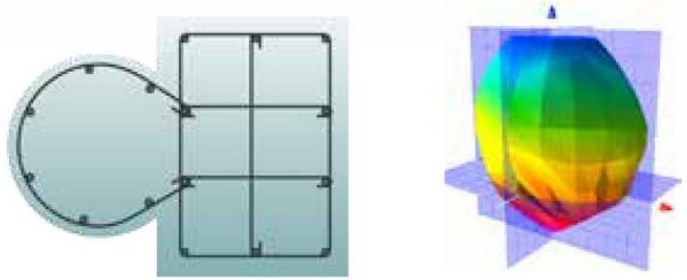
RETROFITTING OF CROSS-SECTIONS



RESPONSE AND DESIGN FOR SHEAR AND TORSION



COMPUTER APPLICATIONS FOR ANALYSIS OF CROSS-SECTIONS



Books & Literature

- Reinforced Concrete - Mechanics and Design (5th ed) by James K. Wright and James G. MacGregor, Pearson-Prentice Hall, Upper Saddle River New Jersey NJ 07458.
- Design of Reinforced Concrete by Jack C. McCormac.
- Design of Concrete Structures (14th Edition), by Arthur H. Nilson, David Darwin, and Charles W. Dolan, McGraw-Hill, New York, NY 10020.
- ACI Building Code Requirement for reinforce concrete structures 318-19.
- Edward G. Nawy (2000): Reinforced Concrete A Fundamental approach, 4th Edition, Prentice-Hall International, Inc.
- Arthur H. Nilson, David Darwin and Charkes W. Dolan (2005): Design of Concrete Structures, 13th Edition, McGraw-Hill.
- Reinforced Concrete Structures by R. Park T. Paulay (2009), ISBN:9780471659174, Copyright © 1975 John Wiley & Sons, Inc.

Books & Literature

- Properties of Concrete, 4th edition by A. M. Neville.
- Concrete by Sidney Mindess [2nd Edn].
- T. Pauley, and M. J. N. Priestley, (1992): Seismic Design of Reinforced Concrete and Masonry Buildings, John Wiley and Sons, New York.
- Bungale S. Taranath (2010): Reinforced Concrete Design of Tall Buildings, Taylor and Francis Group, LLC.

Analysis and Design Software

- ETABS v 2016
- SAP 2000 v 17
- SAFE
- GEAR
- Section Designer
- CSI Bridge
- CSI COL
- PERFORM 3D
- ANSYS
- ABAQUS CAE
- LS DYNA
- DRAIN 3D
- …

Textbooks References and Reading Material

- **International Standards/Guidelines**

1) ASCE 7-16 (2017): Minimum Design Loads and Associated Criteria for Buildings and Other Structures (ASCE/SEI 7-16)
2) ACI CODE-318-19: Building Code Requirements for Structural Concrete and Commentary
3) TBI (2010): Guidelines for Performance-Based Seismic Design of Tall Buildings - PEER
4) FEMA 356 (2000): Pre-standard and Commentary for the Seismic Rehabilitation of Buildings
5) ATC-40 (1996) Seismic Evaluation and Retrofit of Concrete Buildings, USA
6) ASCE/SEI 41-13 (2014): Seismic Rehabilitation of Existing Buildings (ASCE/SEI 41-13)
7) Council on Tall Buildings and Urban-Habitat (2008): Recommendations for the Seismic Design of High-Rise Buildings.
8) PEER/ATC-72-1 (2011): Modeling and Acceptance Criteria for Tall Buildings, USA

Textbooks References and Reading Material

- **Research Journals**

  1) ACI Structural Journal, ACI
  2) Journal of Structural Engineering, ASCE
  3) Engineering Structures, Elsevier
  4) Structural Engineering International Journal, IABSE
  5) Earthquake Engineering & Structural Dynamics, Wiley
  6) The Structural Design of Tall and Special Buildings
  7) Soil Dynamics and Earthquake Engineering, Elsevier
  8) Magazine of Concrete Research, ICE

Marks Distribution & Grading

Assignments	10%
Quizzes	10%
OHT Exams	30%
Complex Engineering Problem	10%
End Semester Exam	40%
Total	100%

Course Websites

https://www.structurespro.info

https://www.fawadnajam.com

Instructor and Contact

- Monday to Friday – Working Hours
- Office No. 14, Ground Floor, PG Wing Building, SCEE, NUST

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Thank You!